Feature Extraction for Noise Robust Digit Recognition

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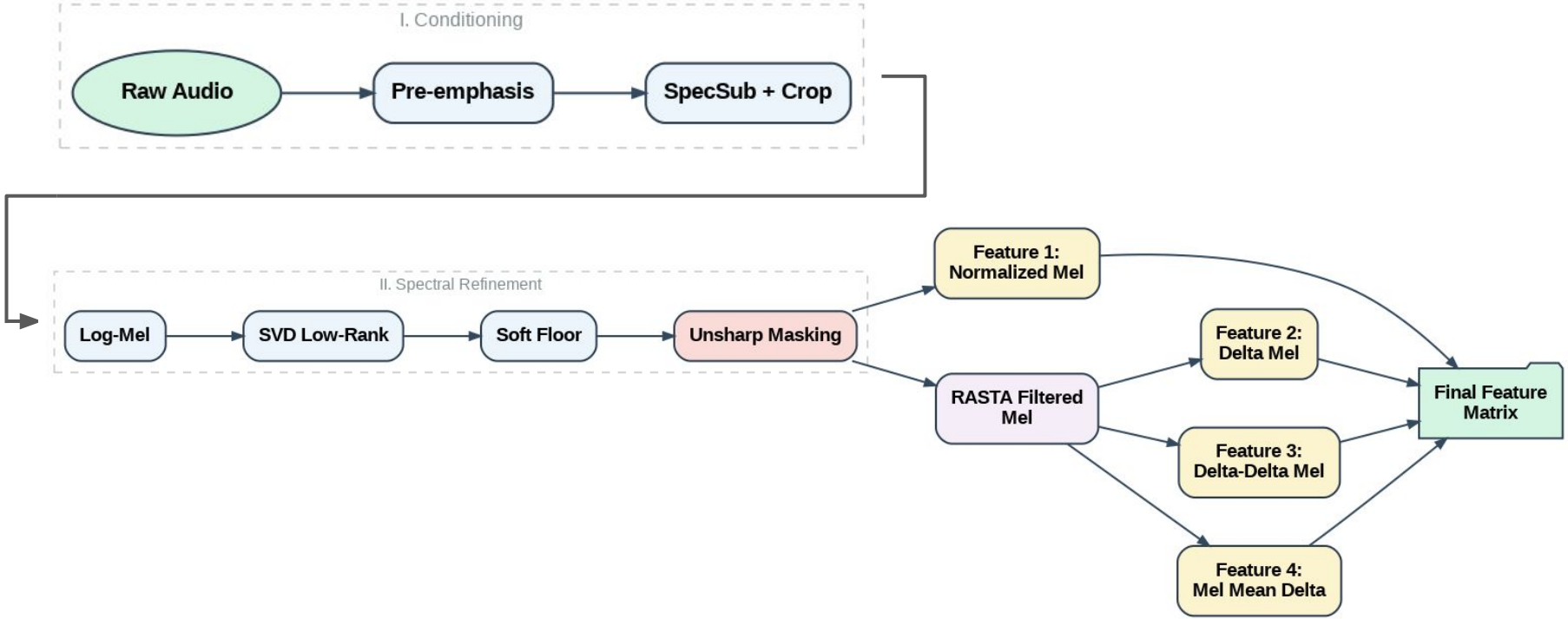
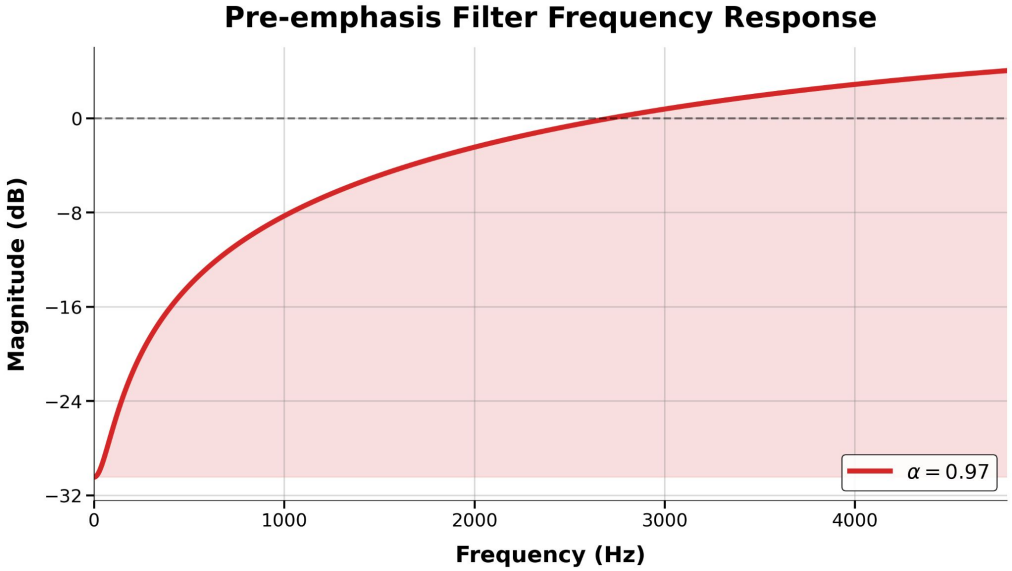
Feature Extraction Pipeline

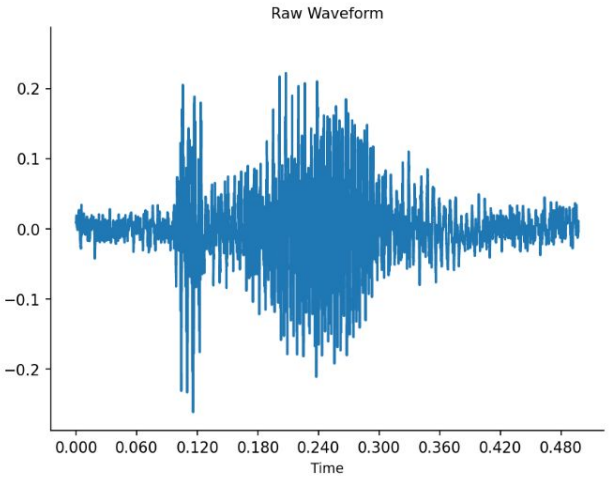


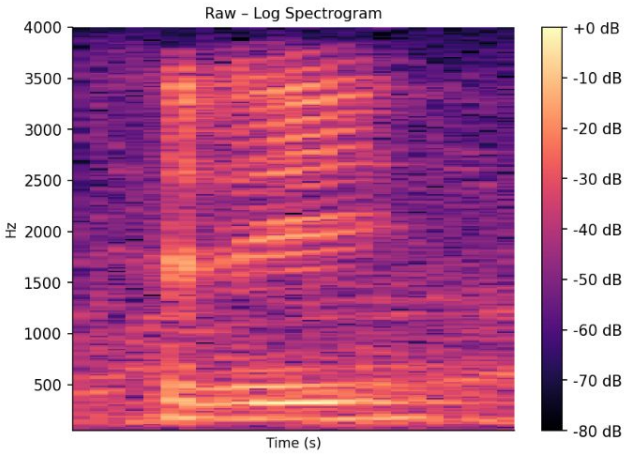
Pre-emphasis (Rationale)

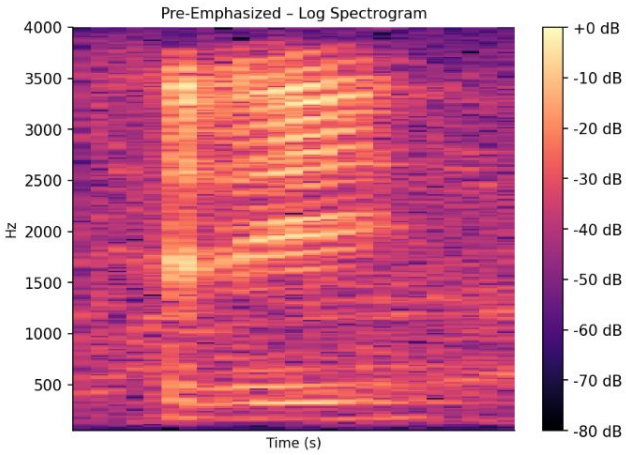
- Human speech has a natural energy decay at higher frequencies [1]
- A high-pass filter balances the energy
- High-frequency consonants (e.g., /s/, /f/, /t/) become more audible
- Higher formants are enhanced



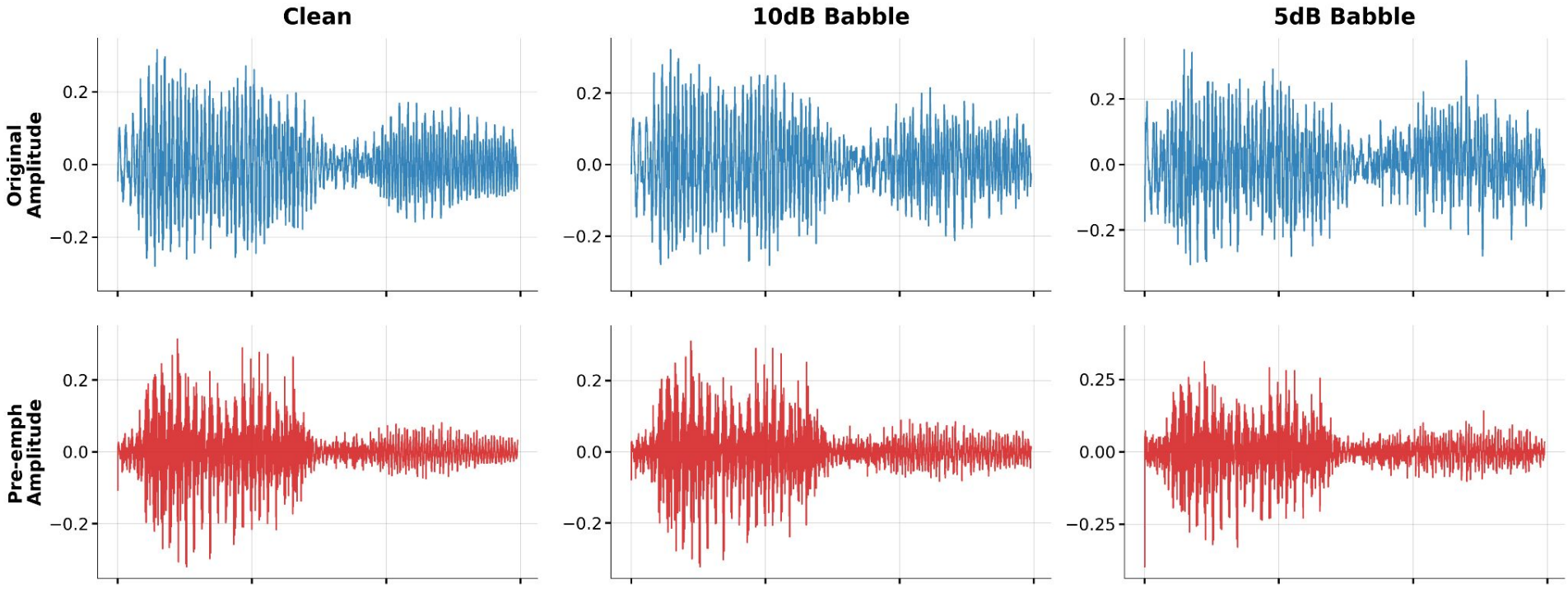
Pre-emphasis (10dB SNR Example)





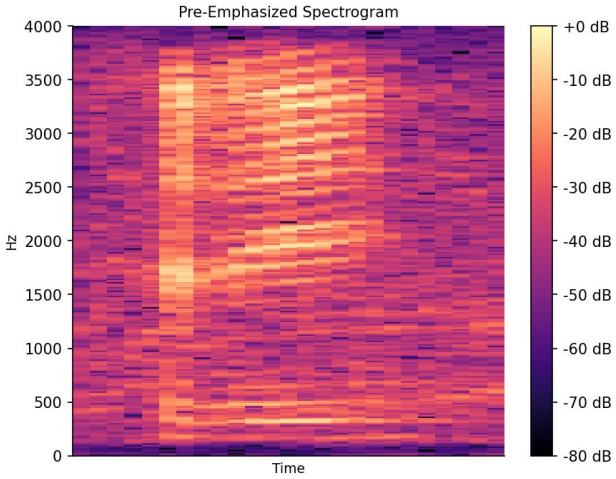


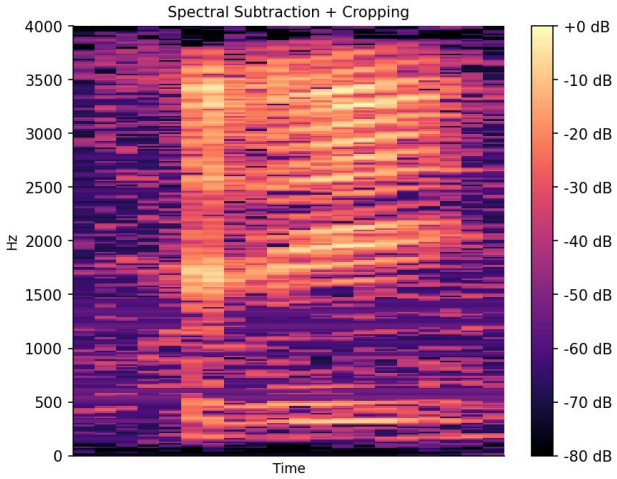
Pre-emphasis (Time Domain Comparison)



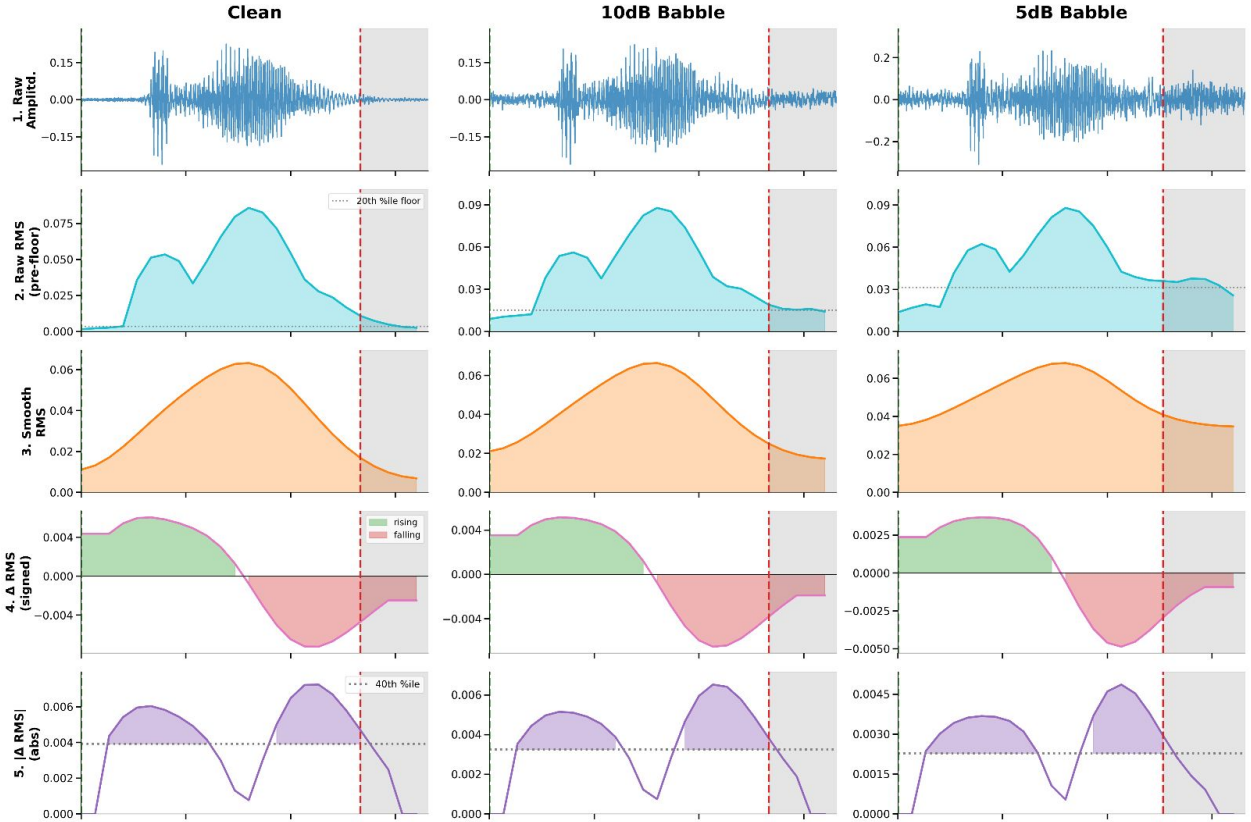
Cropping + Spectral Subtraction (Rationale)

- Cropping: Using VAD (Voice Activity Detection) based on Δ rms energy to detect and crop out the silent part of the audio.
- Spectral Subtraction (noise removal): Find the noise floor from the silent part and subtract it from each frame.

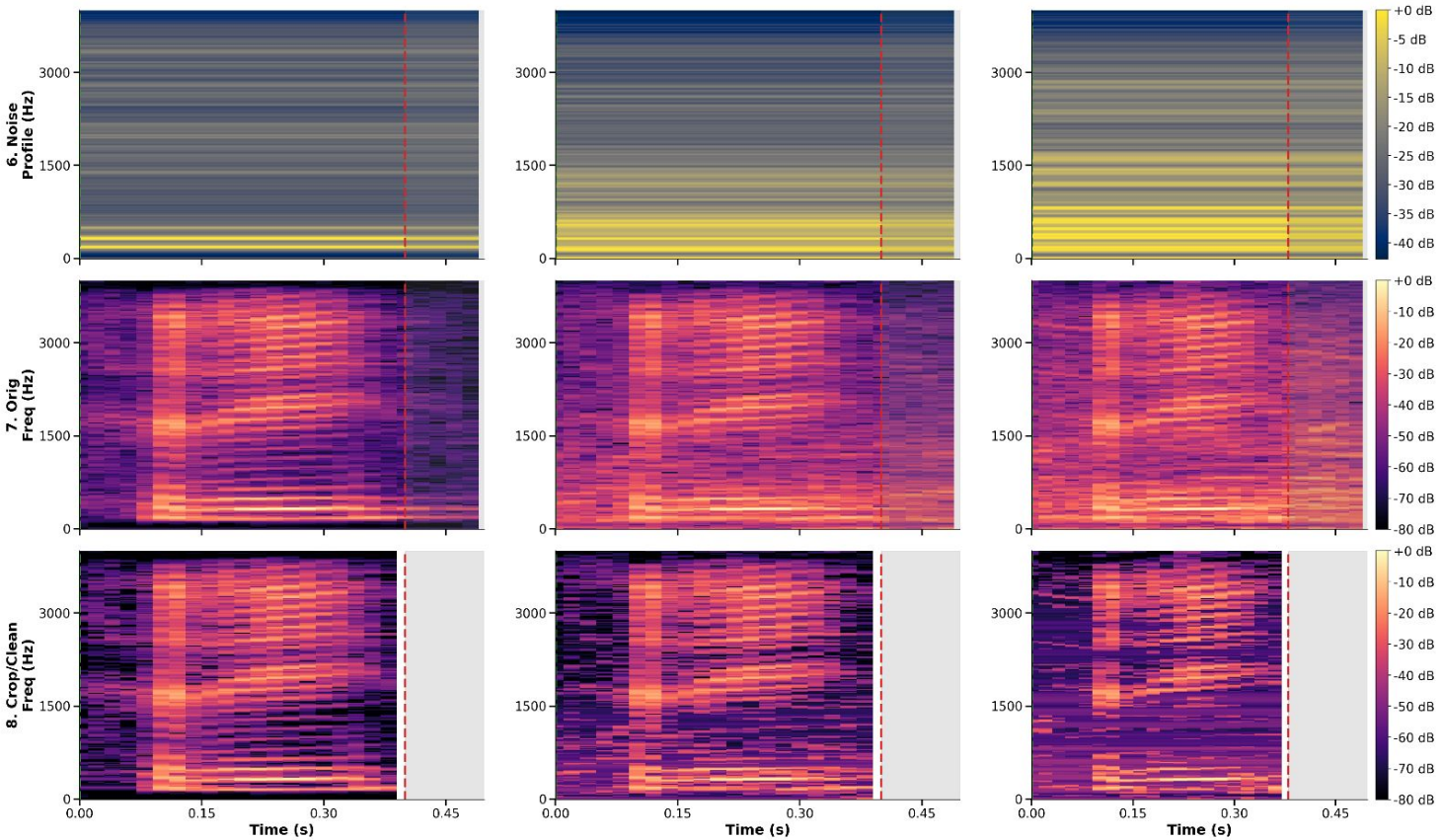




Cropping + Spectral Subtraction by VAD

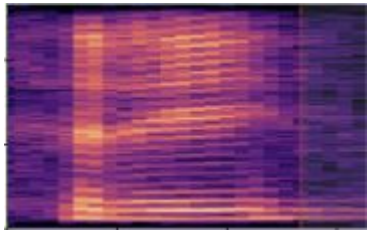


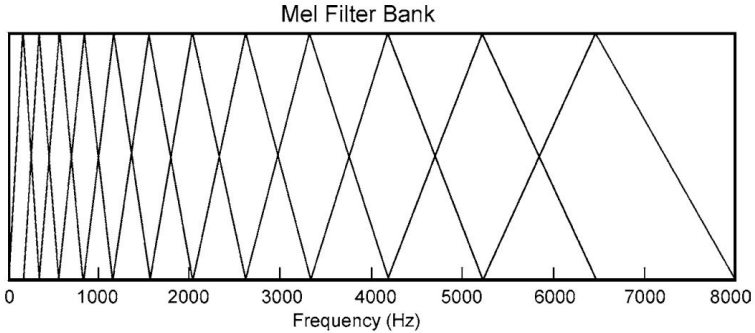
Cropping + Spectral Subtraction by VAD

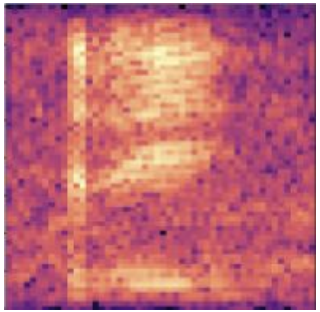


Log-Mel Spectrogram (Rationale)

- Human hearing follows the Mel scale:
  - Compute standard spectrogram
  - Filter with mel filter bank
  - Log to extract power



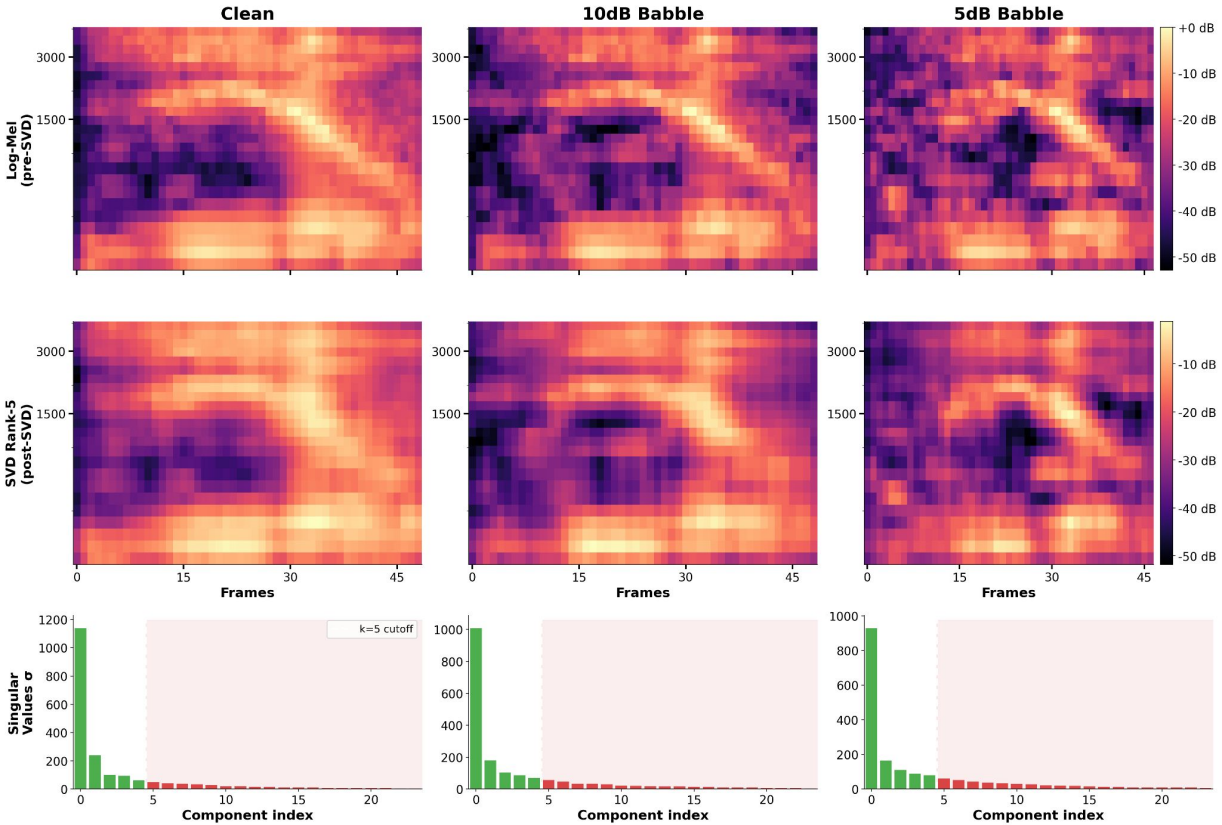




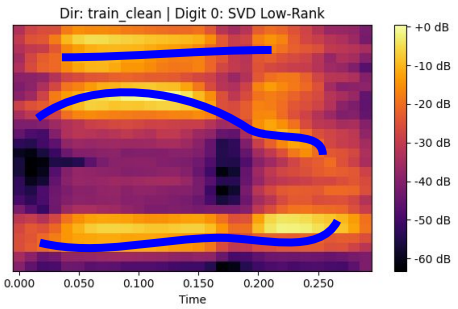
SVD Low-Rank Reconstruction

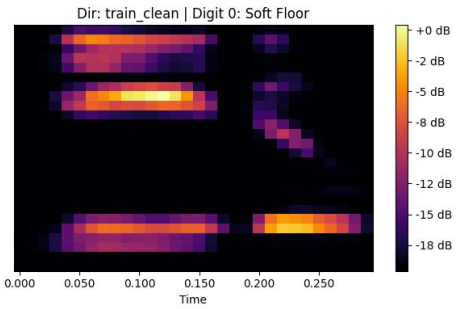
Keep the high variance/energy correlations contained in the auditory features.

Remove the noise which has low spectro-temporal correlation.



Soft Floor (Rationale)





$$\text{Softplus}(x) = \frac{1}{\beta} * \log(1 + \exp(\beta * x))$$

*Original SoftPlus Function [2]

$$f(x, \tau) = \tau + 2\ln\left(1 + e^{2(x-\tau)}\right)$$

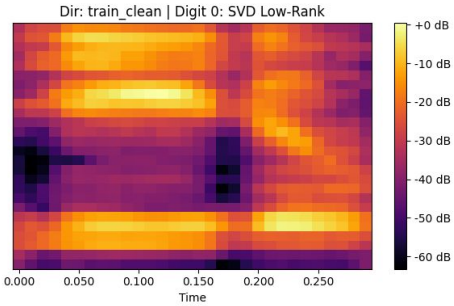
$$\tau = (argmax(x) - 20)$$

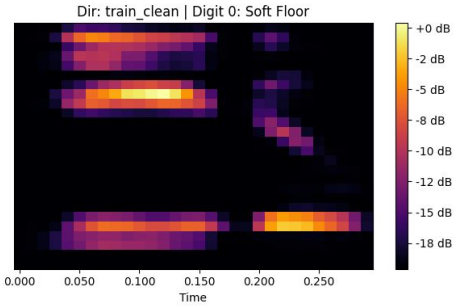
*Aggressive SoftPlus function with dynamic threshold

- Consistent formant extraction via adaptive thresholding
- Similar to LPC: Most of the useful energy is contained in formant peaks

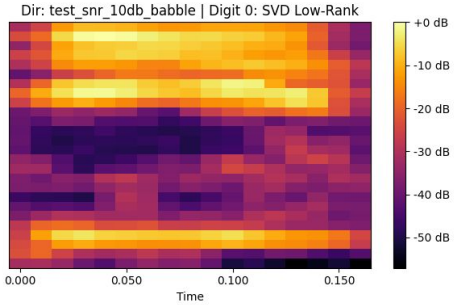
Soft Floor Example

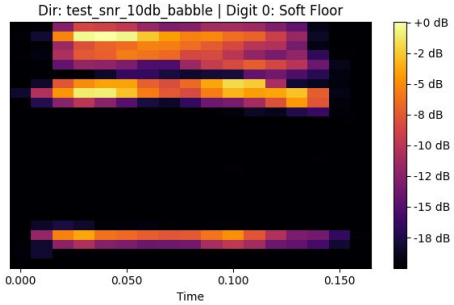
Clean



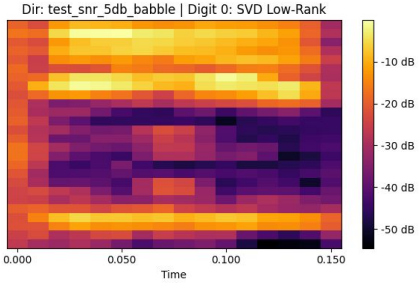


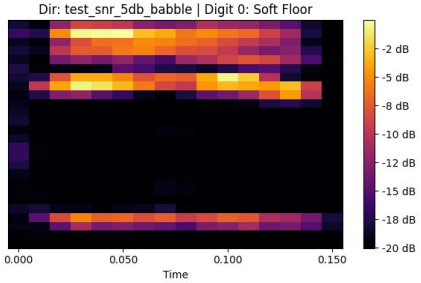
10dB SNR





5dB SNR





Unsharp Masking (Rationale)

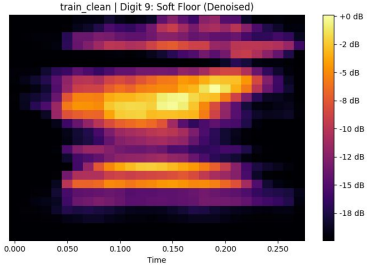
$$U(x) = x + \alpha(x - LPF(x))$$

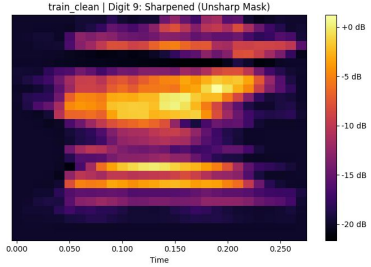
$$LPF = \mathcal{N}(0, 0.6)$$

- Classic image processing technique to increase contrast at edges [3]
- Helps localize and emphasize formant peaks
- Step 1: Subtract smoothed image from itself
- Step 2: Scale the resulting “sharp mask”
- Step 3: Add sharp mask onto original image

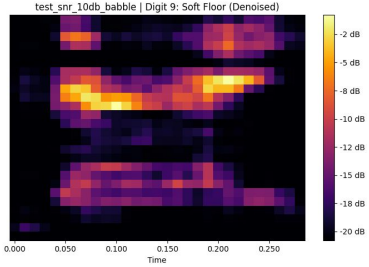
Unsharp Masking Examples

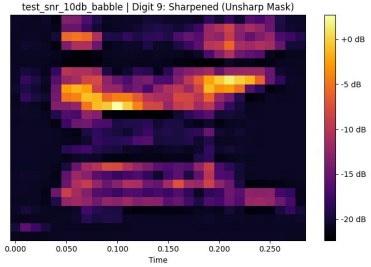
Clean



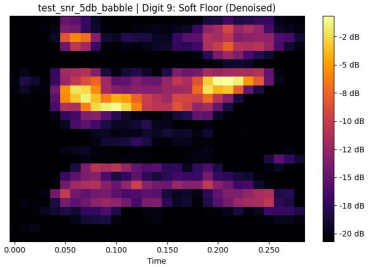


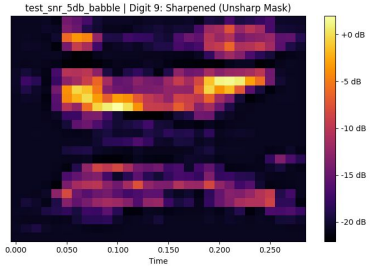
10dB SNR





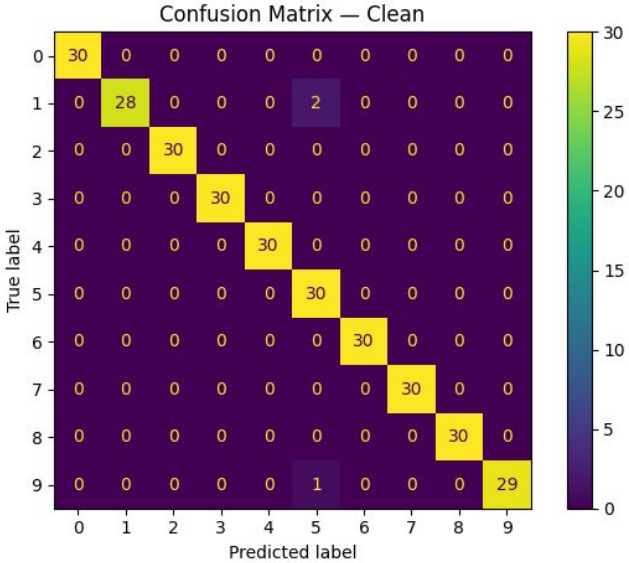
5dB SNR

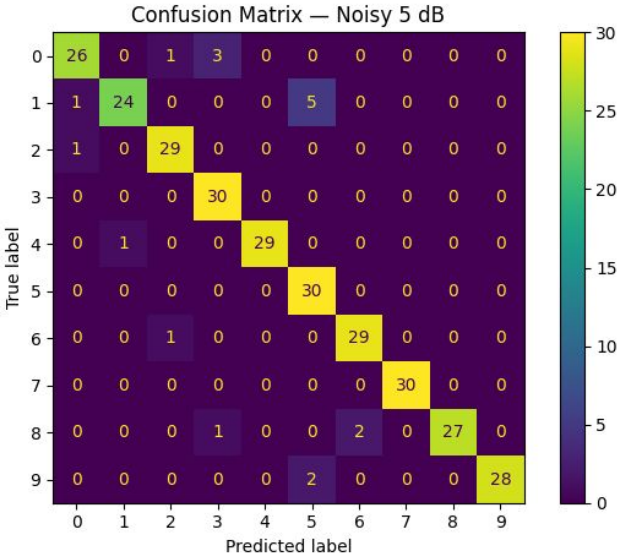


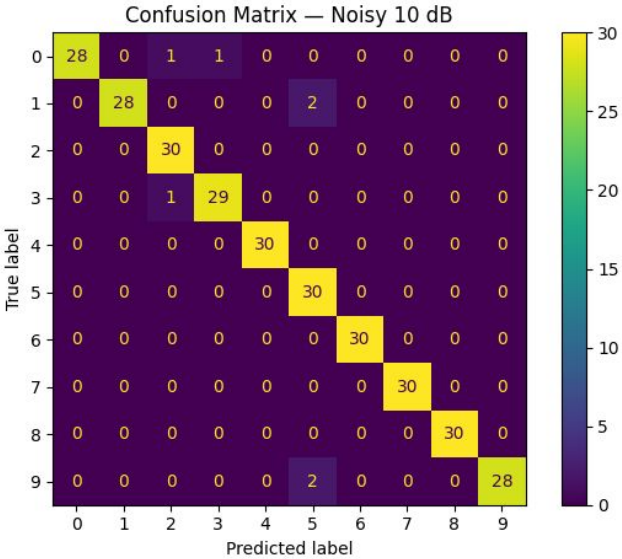


Spectral Refinement - Effectiveness

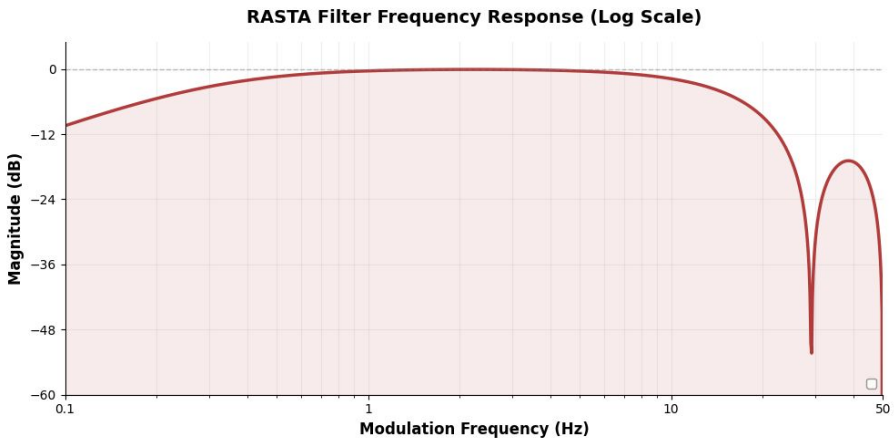
Accuracies	**Clean**	**5dB**	**10dB**
Conditioning (prev. step)	0.99	0.76	0.90
Replace MFCC with Spectral Refinement	0.99	0.94	0.98







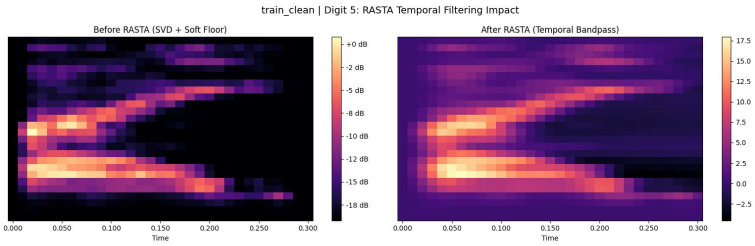
RASTA Filter (Rationale)



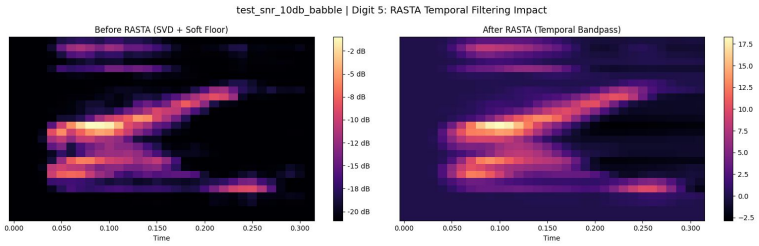
- Before calculating deltas, the Mel-Spectrogram must be smoothed again
- This stabilizes delta values and makes delta features less sparse
- RASTA is a BP filter with cutoffs at around 1Hz and 12Hz to emphasize changes in the typical auditory temporal time-scale [4]
- In this case, RASTA smears out the edges and simultaneously smooths sharp details in the formants

RASTA Filter Examples

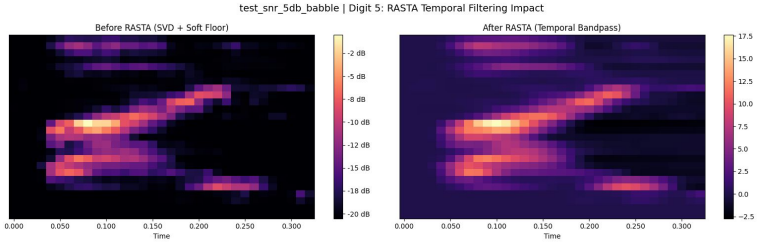
Clean



10dB SNR

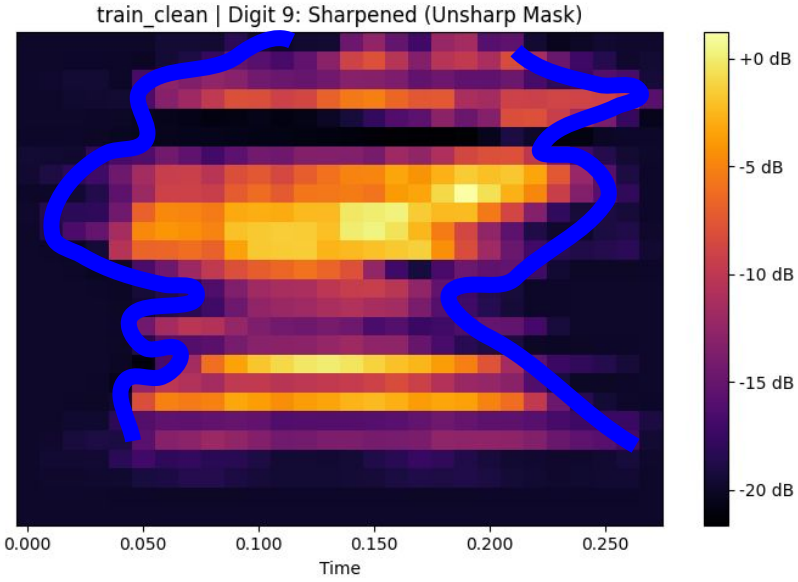


5dB SNR



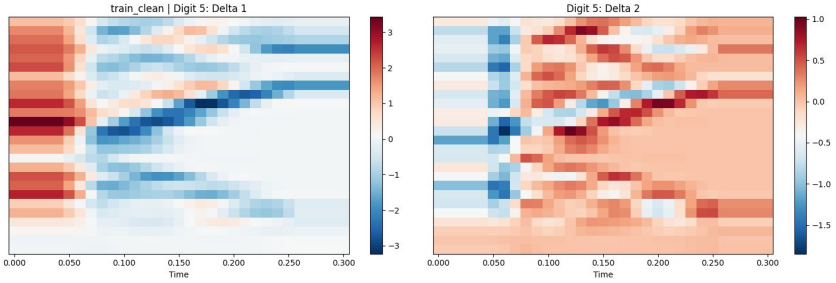
Mel-Spectral Derivatives (Rationale)

- Spectrograms have strong temporal edges
- Derivative helps localize energy
- Each digit has distinct formant “flow” which is captured by 1st and 2nd derivatives

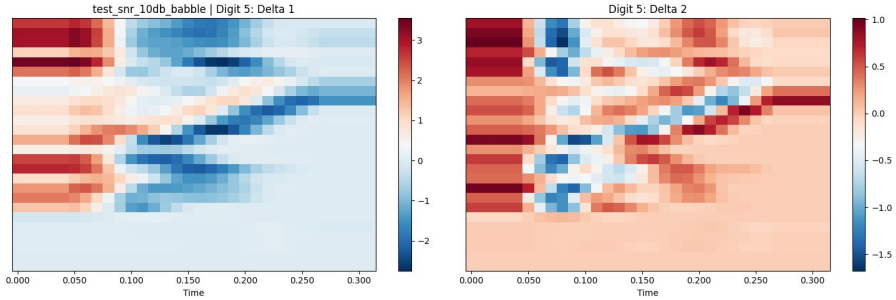


Mel-Spectral Derivative Examples

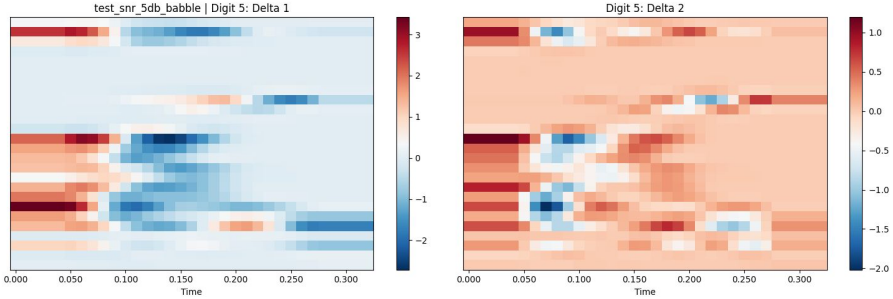
Clean



10dB SNR

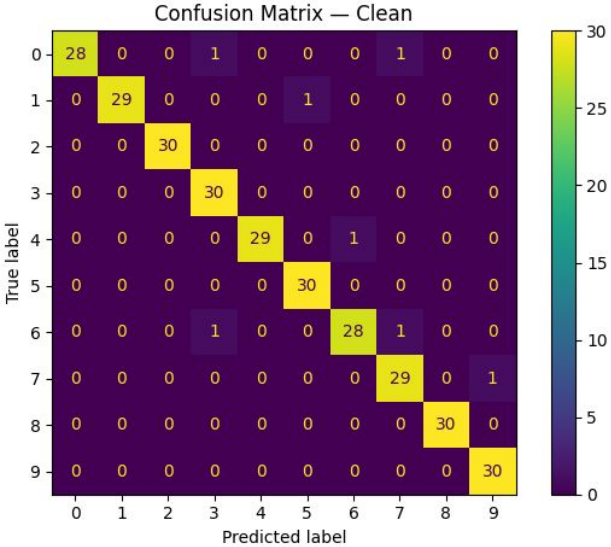


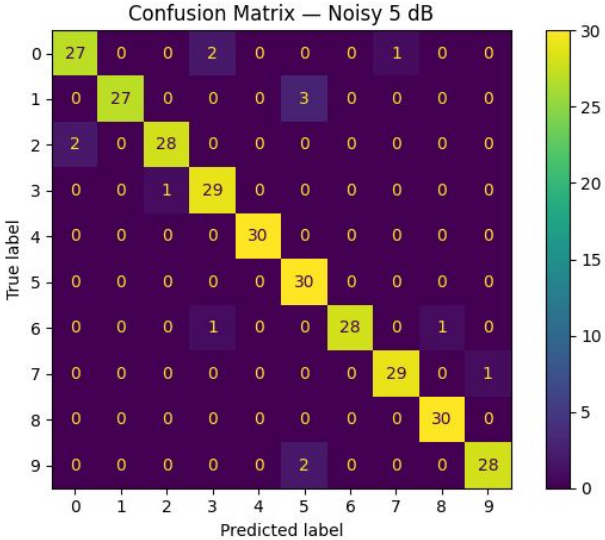
5dB SNR

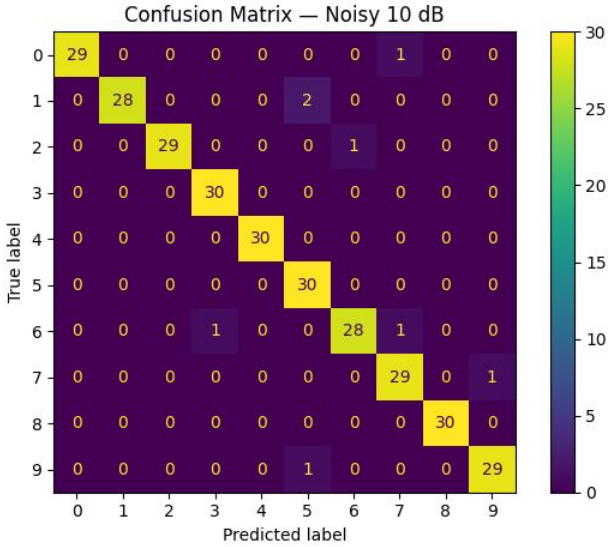


Deltas - Effectiveness

Accuracies	**Clean**	**5dB**	**10dB**
Spectral Refinement (prev. step)	0.99	0.94	0.98
+ Deltas	0.98	0.95	0.97



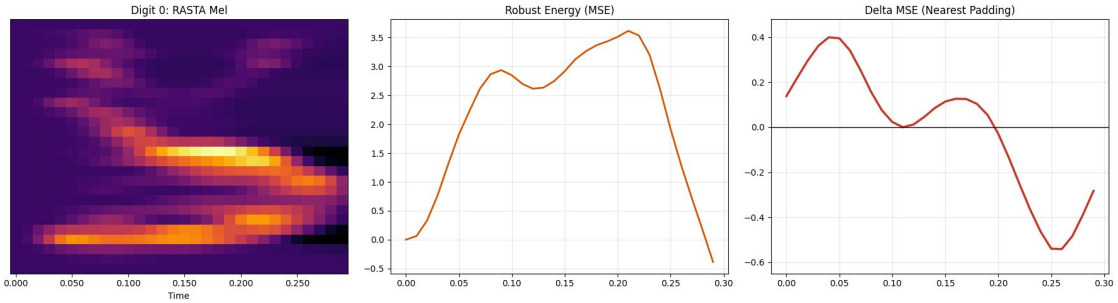


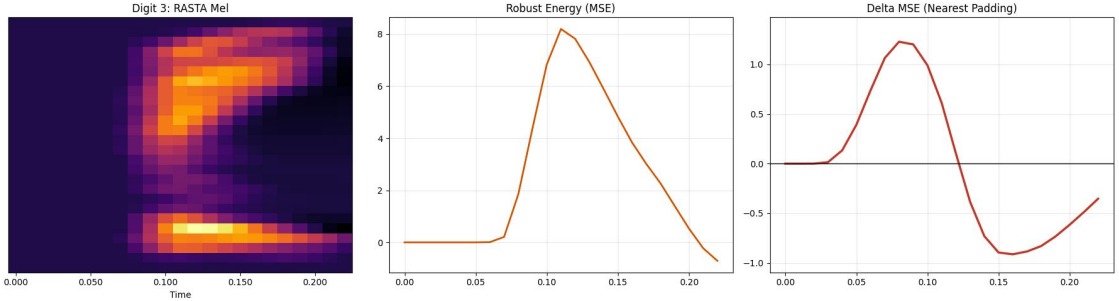


Mean (Energy) Delta Rationale

- Similar to the spectrograms, the energy have strong temporal edges
- Energy highlights the magnitude of the mel-spectrogram over time
- Each digit has distinct energy “flow” which is captured by 1st derivative

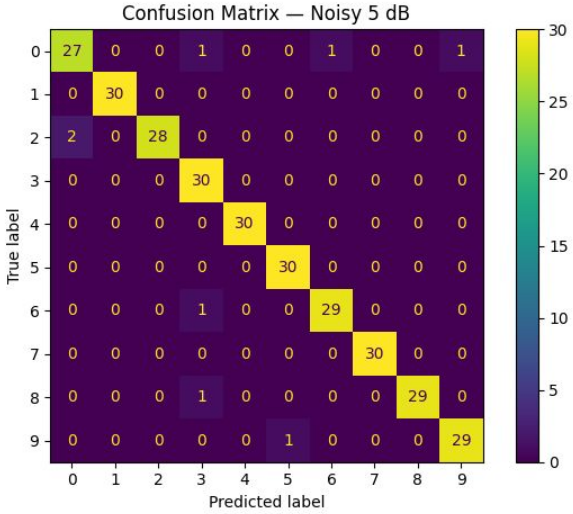
Mean Delta Examples

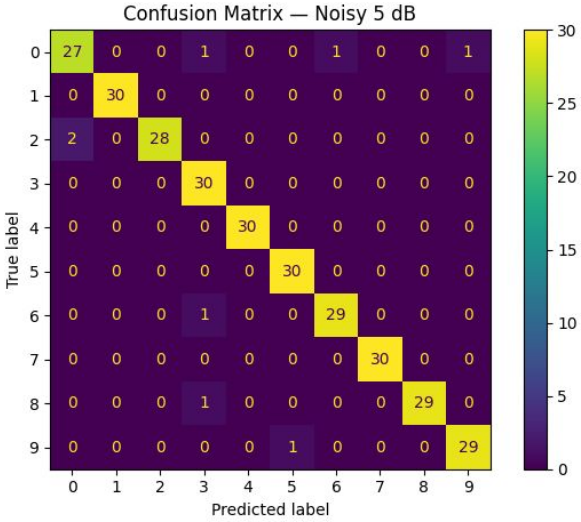


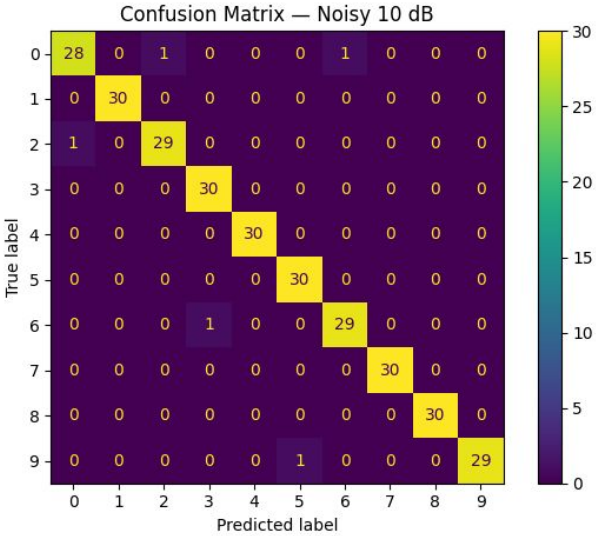


Final Results: ([mels_base, mels_delta, mels_delta2, mean_delta])

Accuracies	**Clean**	**5dB**	**10dB**
Baseline (26,)	99.00%	51.00%	70.00%
Final Feature Set (73,)	98.67%	97.33%	98.33%







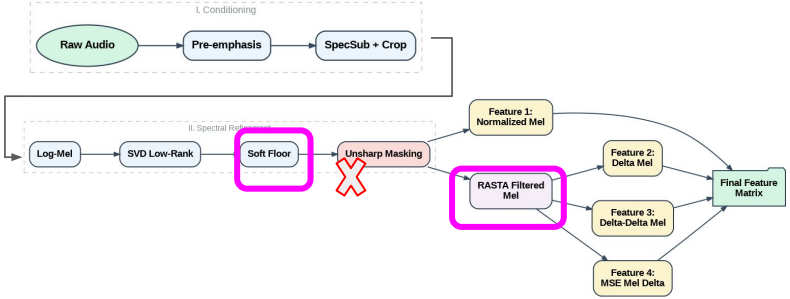
Discussion

Method	**Clean**	**5dB**	**10dB**
Baseline accuracy	99.00%	51.00%	70.00%
Our accuracy	98.67%	97.33%	98.33%
Improvement	-0.33%	+46.33%	+28.33%

- Achieved extreme noise robustness!
- Mel-Spectrogram is a good starting point to extract features
- Intuitive visual and temporal features are a good way to achieve orthogonal input vectors

Future Improvements

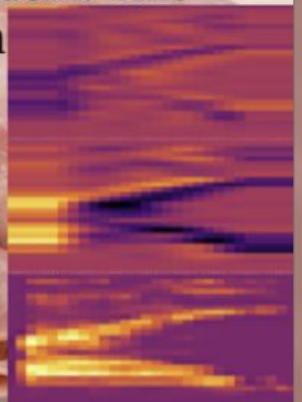
- Reduce redundancy and complexity



- Extract syllable information can narrow down valid selections during classification: (0,7) vs other digits

Questions?

"Is that **speech processed**? If it's **processed** I don't want it"
Ma'am, that is a **24 mel filter slab** of **frequency bins**. It has no **fricatives**, **pitch**, or **coarticulation**. It is the **logarithm** of **heuristically-windowed spectrograms**, **thresholded**, **convolved**, **band-pass filtered**, and ultimately **quantized** in an **unholy SVD low-rank reconstruction**. God had no hand in the **prediction** of this **utterance**. The fact that this **STFT monolith** exists proves that God is either **computally limited** to alter His universe or **invariant** to the **non-linearities** taking place in his kingdom. This **concatenation of features** is more than **lossy data**. It is a physical declaration of mankind's contempt for the **Nyquist-Shannon theorem**.
It is **over-fitted hubris manifest**.
We also have a **lower bitrate variety** if you would prefer that.



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